Automatic Hand Sanitizer Dispenser - Project Report

Introduction
An Automatic Hand Sanitizer Dispenser is a touch-free device that dispenses sanitizer automatically when a hand is placed near the sensor. It helps maintain hygiene by reducing physical contact with the sanitizer bottle. During the COVID-19 pandemic, touch-free sanitizing systems became very important because they reduced the spread of germs and viruses.

This project is simple, low-cost, portable, and easy to use. It is suitable for hospitals, schools, colleges, offices, shopping malls, airports, restaurants, and other public places.

Why We Chose This Project
Maintaining hand hygiene is essential for preventing the spread of infectious diseases. Manual sanitizer bottles require physical contact, which may transfer germs from one user to another.

Components Used
IR Sensor Module, 5V Relay Module, Mini DC Water Pump, 3.7V–5V Battery, Sanitizer Bottle, Connecting Wires.

Working Principle
The IR sensor detects a hand, sends a signal to the relay, which switches on the DC pump to dispense sanitizer. When the hand is removed, the relay switches off the pump, stopping the flow.

Advantages
Touch-free operation, improved hygiene, reduced germ spread, less sanitizer wastage, low cost, portable, easy maintenance.

Applications
Hospitals, Schools, Colleges, Offices, Shopping Malls, Airports, Railway Stations, Hotels, Restaurants, Homes.

Limitations
Affected by direct sunlight, requires battery charging, sensor adjustment, and regular sanitizer refilling.

Future Improvements
Arduino timer control, IoT monitoring, rechargeable battery, sanitizer level indicator, LED/voice indication.

Conclusion
The Automatic Hand Sanitizer Dispenser is a simple, efficient, and cost-effective hygiene device that demonstrates the practical application of automation and electronics in solving real-life problems.